# An $N^{\sqrt{2}+o(1)}$ Upper Bound for Bilinear Sum-of-Squares Composition Formulas

An algebraic consequence of Appendix B of
*A $2^{o(n)}$ Time Algorithm for 3-SAT*

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**Abstract**

Assuming the algebraic identities and uniform rank estimates in Appendix B of the 26 October 2026 Claude/Anthropic manuscript *A $2^{o(n)}$ Time Algorithm for 3-SAT*, its crossing decomposition implies an upper bound of $N^{\sqrt{2}+o(1)}$ for Naslund's crossing-decomposition quantity $P(N)$. It also yields an identity expressing the product of two sums of $N$ squares as a sum of at most $N^{\sqrt{2}+o(1)}$ products of rational bilinear forms. Over the complex numbers, each such product is a sum of two squares of bilinear forms. Consequently, the same hypotheses imply a complex bilinear sum-of-squares composition formula with at most $N^{\sqrt{2}+o(1)}$ squares, improving the $O(N^{1.62})$ bound of Hrubeš (2024, 2026). No modification of the construction in Appendix B is necessary. The implication uses only its algebraic identities and rank estimates, not the SAT algorithm or its running-time analysis.

## 1 Main result

Let $M_{\mathbb{C}}(N)$ denote the minimum integer $M$ for which there exist complex bilinear forms $F_1, \dots, F_M$ satisfying the polynomial identity

$$\left(\sum_{i=1}^{N} a_i^2\right)\left(\sum_{j=1}^{N} b_j^2\right) = \sum_{\ell=1}^{M} F_\ell(a,b)^2.$$

Here each form is homogeneous of degree one in $a$ and of degree one in $b$. The squares are algebraic squares, without complex conjugation.

**Theorem 1** (Conditional on the Appendix B construction)**.** *Assume the identities and the uniform rank-factorization estimate* (1)–(4) *below, established in Appendix B of* A $2^{o(n)}$ Time Algorithm for 3-SAT *[1]. There is an absolute constant $C$ such that, for every $N \geq 2$,*

$$M_{\mathbb{C}}(N) = O\Big(N^{\sqrt{2}}(\log N)^C\Big).$$

*In particular, $M_{\mathbb{C}}(N) \leq N^{\sqrt{2}+o(1)}$. The same upper bound holds for the number of rational bilinear products required to express the product of two sums of squares, and for the crossing-decomposition quantity $P(N)$ defined in Appendix G of that manuscript.*

## 2 The crossing decomposition

First suppose $N = 2^\lambda$, and identify $[N] = \{1, \dots, N\}$ with $\{0,1\}^\lambda$. Define the crossing tensor

$$D(x,y,u,v) = [x = v][y = u],$$

where brackets denote indicators.

Appendix B, Theorem B, constructs an integer $h = O(\lambda^2)$, integer-valued functions

$$A_s(x, y, u, v), \qquad -2h \le s \le 2h,$$

and a nonzero integer $d = h!$ such that

$$dD(x, y, u, v) = \sum_{s=-2h}^{2h} A_s(x, y, u, v). \tag{1}$$

Moreover, each $A_s$ admits two factorizations. The first separates $(x, y)$ from $(u, v)$:

$$A_s(x, y, u, v) = \sum_{q=1}^{R_s} f_{s,q}(x, y) g_{s,q}(u, v). \tag{2}$$

The second separates $(x, u)$ from $(y, v)$:

$$A_s(x, y, u, v) = \sum_{q=1}^{C_s} \widetilde{f}_{s,q}(x, u) \widetilde{g}_{s,q}(y, v). \tag{3}$$

The factors are integer-valued, and the positive factorization lengths satisfy

$$\log_2 R_s + \log_2 C_s \le 2\sqrt{2}\,\lambda + O(\log(2\lambda))\,, \tag{4}$$

uniformly in $s$. This follows from the bound on the rounded-up binary logarithms in Appendix B, equation (56), derived using the Hardy–Ramanujan asymptotics for integer partitions. The $O(\log(2\lambda))$ term agrees with $O(\log \lambda)$ asymptotically and also covers $\lambda = 1$.

## 3 Choosing the cheaper factorization

For each $s$, use whichever of (2) and (3) contains fewer terms. By (4),

$$\min(R_s, C_s) \le \sqrt{R_s C_s} \le 2^{\sqrt{2}\lambda + O(\log(2\lambda))}.$$

Since $N = 2^\lambda$, this gives

$$\min(R_s, C_s) \le N^{\sqrt{2}} (\log N)^{O(1)}. \tag{5}$$

There are $4h+1 = O(\lambda^2)$ values of $s$. Expanding each $A_s$ using its cheaper orientation and dividing (1) by $d$ therefore expresses $D$ as a sum of $L$ rational rank-one functions, each separated in one of the two orientations, where

$$L = O\Big(N^{\sqrt{2}} (\log N)^C\Big) \tag{6}$$

for an absolute constant $C$. By the definition in Appendix G, this proves

$$P(N) \le L \le N^{\sqrt{2}+o(1)}. \tag{7}$$

## 4 From crossing factorizations to bilinear products

Multiply (1), after division by $d$, by $a_x b_y a_v b_u$, and sum over all $x, y, u, v \in [N]$. The left-hand side becomes

$$\sum_{x,y,u,v} [x = v][y = u]\, a_x b_y a_v b_u = \left(\sum_x a_x^2\right)\left(\sum_y b_y^2\right). \tag{8}$$

For a rank-one term in orientation (2), namely $f(x,y)g(u,v)$, the corresponding contribution, before normalization by $d$, is

$$\left(\sum_{x,y} f(x,y)a_x b_y\right)\left(\sum_{u,v} g(u,v)b_u a_v\right),$$

a product of two bilinear forms. For a rank-one term in orientation (3), namely $\widetilde{f}(x,u)\widetilde{g}(y,v)$, the contribution is

$$\left(\sum_{x,u} \widetilde{f}(x,u)a_x b_u\right)\left(\sum_{y,v} \widetilde{g}(y,v)b_y a_v\right).$$

This is again a product of two bilinear forms.

Hence (1) gives an identity

$$\boxed{\left(\sum_{i=1}^{N} a_i^2\right)\left(\sum_{j=1}^{N} b_j^2\right) = \sum_{q=1}^{L} F_q(a,b)G_q(a,b),} \tag{9}$$

where all $F_q, G_q$ are rational bilinear forms and $L \le N^{\sqrt{2}+o(1)}$. The normalization factor $1/h!$ is absorbed into one factor of each product.

## 5 Conversion to complex sums of squares

For any bilinear forms $F, G$ over $\mathbb{Q}$, the identity

$$FG = \left(\frac{F+G}{2}\right)^2 + \left(\frac{\mathrm{i}(F-G)}{2}\right)^2 \tag{10}$$

holds over $\mathbb{C}$, where $\mathrm{i}^2 = -1$. Both expressions inside the squares remain bilinear forms in $a, b$.

Applying (10) to every term of (9) produces a sum of at most $2L$ squares of complex bilinear forms. Thus

$$M_{\mathbb{C}}(N) \le 2L = O\Big(N^{\sqrt{2}}(\log N)^C\Big).$$

In fact, the coefficients produced lie in $\mathbb{Q}(\mathrm{i})$.

For arbitrary $N$, choose $\lambda = \lceil \log_2 N \rceil$, apply the construction to $2^\lambda$ variables, and set the excess variables to zero. Since $N \le 2^\lambda < 2N$, the same asymptotic bound follows. Restricting all four tensor indices to $[N]$ likewise gives the bound for $P(N)$. This proves Theorem 1.

## 6 Comparison with prior work

Hrubeš established an $O(N^{1.62})$ upper bound for complex bilinear sum-of-squares composition formulas in *A Subquadratic Upper Bound on Sum-Of-Squares Composition Formulas* (CCC 2024) [2], subsequently developed in *A Subquadratic Upper Bound on Hurwitz's Problem and Related Noncommutative Polynomials* (2026) [3]. Related connections to noncommutative circuits appear in [4].

The construction above gives

$$\sqrt{2} \approx 1.41421356 < 1.62,$$

so, under the stated hypotheses, it improves the exponent by approximately 0.206.

This conclusion is conditional only on the correctness of the algebraic identities and uniform rank-factorization estimates in the Claude/Anthropic manuscript. It does not rely on that

manuscript's SAT algorithm, recursive running-time analysis, or formalization of the Exponential Time Hypothesis (ETH). This note proves the implication from the stated algebraic hypotheses; it does not independently establish those hypotheses or the claimed SAT running time.

## 7 Remarks and attribution

**Rational versus complex coefficients.** Equation (9) holds over $\mathbb{Q}$, as the Appendix B factors are integer-valued before normalization. Equation (10) uses $\mathrm{i} = \sqrt{-1}$, so the asserted upper bound for sums of squares is over $\mathbb{C}$. It does not establish a sum-of-squares identity of the same size over $\mathbb{R}$ or $\mathbb{Q}$.

**Relation to Naslund's problem.** Equation (7) strengthens, under the stated hypotheses, the upper bound for $P(N)$ discussed in Appendix G. Naslund's partition-rank work is [5]. The conversion to bilinear products and then complex squares establishes a separate consequence for Hurwitz's composition problem.

**Attribution.** The construction is that of Appendix B of Claude's manuscript for Anthropic. The observation here is that its two-sided factorization estimate implies a stronger bound for $P(N)$ and for complex sum-of-squares composition formulas.

## References


[1] Claude (Anthropic). *A $2^{o(n)}$ Time Algorithm for 3-SAT.* Manuscript dated 26 October 2026. Appendix B, Theorem B, equations (53)–(56); Appendix G. Version in `anthropics/formal-math`, commit `16a5ce4`.

[2] Pavel Hrubeš. *A Subquadratic Upper Bound on Sum-Of-Squares Composition Formulas.* In *39th Computational Complexity Conference (CCC 2024)*, LIPIcs 300, Article 12, pp. 12:1–12:11, 2024. doi:10.4230/LIPIcs.CCC.2024.12.

[3] Pavel Hrubeš. *A Subquadratic Upper Bound on Hurwitz's Problem and Related Noncommutative Polynomials. SIAM Journal on Computing* 55(2), 451–467, 2026. doi:10.1137/24M1666884.

[4] Pavel Hrubeš, Avi Wigderson, and Amir Yehudayoff. *Non-commutative Circuits and the Sum of Squares Problem. Journal of the American Mathematical Society* 24(3), 871–898, 2011. doi:10.1090/S0894-0347-2011-00694-2.

[5] Eric Naslund. *The Partition Rank of a Tensor and k-Right Corners in $\mathbb{F}_q^n$. Journal of Combinatorial Theory, Series A* 174, Article 105190, 2020. doi:10.1016/j.jcta.2019.105190.